Stats 506 F23 - Answer Key

Worked reference answers. Match the year and problem label before checking your work. Minor errata are clarified where relevant.

https://dept.stat.lsa.umich.edu/~jerrick/courses/stat506_f23/midterm-solns.html

Problem 1

a.

q = 25. The selected odd values are 1, 3, 5, 7, 9.

b.

q = 42. The recycled multipliers are 2, 1, 2, 1, 2; the first and fifth products are 2 and 40. A recycling warning does not prevent the calculation.

c.

```
c(-3, 0, -3)
```

Row sums are 3, 5, 3; column sums are 6, 5. Recycle the column-sum vector as 6, 5, 6 before subtraction.

d.

q = 1.25. Trace 10 -> 5 -> 2.5 -> 1.25, then return the value at or below 2.

e.

q = 3. The variable d equals a. Its means are 2 in group b=1 and 3 in group b=2; descending order selects 3.

f.

q = 35. Add successive triangular totals 1 + 3 + 6 + 10 + 15. The next while condition then fails.

Problem 2 - SAS

These questions belong to the F23 scope. SAS is outside the published F26 Lecture 1-9 scope.

a.

```
proc sort data=work.housing;
    by age;
run;
```

b.

```
data work.housing;
    set work.housing;
    birthyear = year - age;
run;
```

c.

```
proc sort data=work.housing;
    by owns_home;
run;
proc means data=work.housing;
    var birthyear;
    by owns_home;
run;
```

Alternatively, use CLASS owns_home with PROC MEANS if no BY processing is required.

Problem 3 - Model translation and Stata

a.

```
lm(y ~ x1 + factor(x2) + x3*x4, data = dat)
```

b.

```
lm(y ~ factor(x1)*factor(x2)*factor(x3) - 1, data = dat)
```

In the Stata interaction specification, the unprefixed x3 is treated as categorical. Remove the intercept.

c.

```
lm(y ~ x1 + x1:x2, data = dat)
```

d.

```
lm(y ~ I(x1^2) + x2 + log(x3), data = dat)
```

e.

```
foreach var of varlist bin1-bin20 {
    replace `var' = `var' - 1
}
```

Subtract one from each 1/2-coded variable.

f.

```
margins race, at(age = (20(10)80))
marginsplot
```

g.

```
generate sd = (x1 - x2)^2
collapse (mean) sd, by(group)
```

h.

```
Q = max(diagonal(X))
```

Insert this line inside the provided Mata block before st_matrix("q", Q).

Problem 4 - Regex matching

Patterns are shown as R string contents. X means a match; a blank means no match.

```
a. "[Aa]*[a-zA-Z]+.{2}o$"
b. "^..?...$"
c. "^[^AS]..?[aeiou]"
d. "(.).\\1"
```

String	a	b	c	d
Adedayo	X			X
Dinah		X	X	
Frida		X	X	
Liam		X	X	
Marlyn				
Milko	X	X		
Mirka		X		
Rupert			X	
Rustik				
Sebastian				

a: A suffix ending in o with enough preceding characters. The optional [Aa]* can match zero characters; it does not forbid an initial A or a.

b: Exactly four or five characters. c: The first character is neither A nor S, and a vowel occurs in position 3 or 4. d: The same character occurs with exactly one intervening character, as in ded.

Problem 5 - SQL output

a

x	y	q	z
9	8	1	2
1	8	2	5

b

x	x1
2	1
9	8

c

q	xx
2	11

d

x	p
2	4
9	NA
7	NA
3	NA
1	NA
6	NA

e

x	p
2	4
NA	8

Problem 6 - Von Neumann extractor

a.

```
vonneumann <- function(p) {
  repeat {
    flips <- rbinom(2, size = 1, prob = p)
    if (flips[1] != flips[2]) return(flips[1])
  }
}
```

The actual R function requires size=1 for a Bernoulli flip. Ignore identical pairs and return the first result from a differing pair.

b.

```
set.seed(506)
p <- 0.7
reps <- 10000
out <- sapply(seq_len(reps), function(i) vonneumann(p))
mean(out)
```

The estimate should be near 0.5 and will vary with the random stream. Both HT and TH have probability p(1-p), so conditional on a differing pair the first result is fair.